Linearized Kinematic Bicycle Model

Chunk C2 – F1TENTH Linear Time-Varying MPC

State $\boldsymbol{z} = [x, y, v, \psi]^T$, input $\boldsymbol{u} = [a, \delta]^T$

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1 Nonlinear continuous-time model

For wheelbase $L > 0$, longitudinal acceleration a, and front steering angle δ,

$$\dot{\boldsymbol{z}} = f(\boldsymbol{z}, \boldsymbol{u}) = \begin{bmatrix} v\cos\psi \\ v\sin\psi \\ a \\ \dfrac{v}{L}\tan\delta \end{bmatrix}. \tag{1}$$

The derivation assumes no tire slip, steering angle strictly inside $(-\pi/2, \pi/2)$, and piecewise-constant inputs over one sample.

2 First-order affine linearization

Let the complete operating point be

$$\bar{\boldsymbol{z}} = [\bar{x}, \bar{y}, \bar{v}, \bar{\psi}]^T, \qquad \bar{\boldsymbol{u}} = [\bar{a}, \bar{\delta}]^T.$$

A first-order Taylor expansion gives

$$\begin{aligned} f(\boldsymbol{z}, \boldsymbol{u}) &\approx f(\bar{\boldsymbol{z}}, \bar{\boldsymbol{u}}) + A_c(\boldsymbol{z} - \bar{\boldsymbol{z}}) + B_c(\boldsymbol{u} - \bar{\boldsymbol{u}}) \qquad (2) \\ &= A_c\boldsymbol{z} + B_c\boldsymbol{u} + \boldsymbol{g}_c, \qquad (3) \end{aligned}$$

where $\boldsymbol{g}_c = f(\bar{\boldsymbol{z}}, \bar{\boldsymbol{u}}) - A_c\bar{\boldsymbol{z}} - B_c\bar{\boldsymbol{u}}$.

The state Jacobian is

$$A_c = \left.\frac{\partial f}{\partial \boldsymbol{z}}\right|_{(\bar{\boldsymbol{z}}, \bar{\boldsymbol{u}})} = \begin{bmatrix} 0 & 0 & \cos\bar{\psi} & -\bar{v}\sin\bar{\psi} \\ 0 & 0 & \sin\bar{\psi} & \bar{v}\cos\bar{\psi} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \dfrac{\tan\bar{\delta}}{L} & 0 \end{bmatrix}. \tag{4}$$

The input Jacobian is

$$B_c = \left.\frac{\partial f}{\partial \boldsymbol{u}}\right|_{(\bar{\boldsymbol{z}}, \bar{\boldsymbol{u}})} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & \dfrac{\bar{v}}{L\cos^2\bar{\delta}} \end{bmatrix}. \tag{5}$$

The continuous affine offset simplifies to

$$\boldsymbol{g}_c = \begin{bmatrix} \bar{v}\bar{\psi}\sin\bar{\psi} \\ -\bar{v}\bar{\psi}\cos\bar{\psi} \\ 0 \\ -\dfrac{\bar{v}\bar{\delta}}{L\cos^2\bar{\delta}} \end{bmatrix}. \tag{6}$$

Although $\bar{x}$, $\bar{y}$, and $\bar{a}$ do not appear after simplification, the Jacobians are still evaluated at the full state-input operating point.

3 Forward-Euler discrete affine model

With sampling period Δt,

$$\boxed{\boldsymbol{z}_{k+1} = A_k\boldsymbol{z}_k + B_k\boldsymbol{u}_k + \boldsymbol{C}_k} \tag{7}$$

where

$$A_k = I + \Delta t A_c, \qquad B_k = \Delta t B_c, \qquad \boldsymbol{C}_k = \Delta t \boldsymbol{g}_c. \tag{8}$$

Therefore,

$$A_k = \begin{bmatrix} 1 & 0 & \Delta t \cos\bar{\psi} & -\Delta t \bar{v} \sin\bar{\psi} \\ 0 & 1 & \Delta t \sin\bar{\psi} & \Delta t \bar{v} \cos\bar{\psi} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \dfrac{\Delta t \tan\bar{\delta}}{L} & 1 \end{bmatrix}, \tag{9}$$

$$B_k = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \Delta t & 0 \\ 0 & \dfrac{\Delta t \bar{v}}{L \cos^2\bar{\delta}} \end{bmatrix}, \quad \boldsymbol{C}_k = \Delta t \begin{bmatrix} \bar{v}\bar{\psi} \sin\bar{\psi} \\ -\bar{v}\bar{\psi} \cos\bar{\psi} \\ 0 \\ -\dfrac{\bar{v}\bar{\delta}}{L \cos^2\bar{\delta}} \end{bmatrix}. \tag{10}$$

4 Consistency check

At the operating point, the affine approximation must reproduce one Euler step of the nonlinear model:

$$A_k \bar{\boldsymbol{z}} + B_k \bar{\boldsymbol{u}} + \boldsymbol{C}_k = \bar{\boldsymbol{z}} + \Delta t\, f(\bar{\boldsymbol{z}}, \bar{\boldsymbol{u}}). \tag{11}$$

This identity is a direct numerical unit test for the implementation. The affine offset is essential: omitting it generally breaks the identity whenever $\bar{\psi} \neq 0$ or $\bar{\delta} \neq 0$.

5 Use inside linear time-varying MPC

For each prediction step k, evaluate $(A_k, B_k, \boldsymbol{C}_k)$ at a nominal pair $(\bar{\boldsymbol{z}}_k, \bar{\boldsymbol{u}}_k)$. The matrices remain fixed during one QP solve, so its dynamics constraints are affine. After applying the first optimized input, update the state and nominal trajectory, re-linearize, and solve again.

This is linear time-varying MPC, not nonlinear MPC: the original trigonometric equations are never optimization constraints.

6 Implementation contract

Quantity	Ordering	Shape
$\boldsymbol{z}_k$	$[x, y, v, \psi]^T$	4×1
$\boldsymbol{u}_k$	$[a, \delta]^T$	2×1
A_k	state transition	4×4
B_k	input transition	4×2
$\boldsymbol{C}_k$	affine offset	4×1

The earlier forward simulator uses $[x, y, \psi, v]$. Convert ordering only at that module boundary; do not mix conventions inside the MPC.